

Maximilian Van Zandt

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EDUCATION

University of California - Berkeley

Berkeley, CA

Master of Arts (M.A.) in Statistics & Data Science

Expected May 2027

- Notable Coursework: Advanced Causal Inference; Advanced Probability; Advanced Statistics; ML for Decision-Making

University of Virginia

Charlottesville, VA

Bachelor of Science (B.S.) in Statistics - Philosophy Minor

May 2026

- GPA: 3.91/4.00 - Dean's List
- Notable Coursework: Linear Models; Multivariate Statistics; Nonparametric Statistics; Data Mining; Machine Learning; Time Series; Clinical Trials Methodology

SKILLS

- **Statistical Methods:** Causal Inference, Experimental Design (RCT, Power Analysis), Hypothesis Testing, A/B/n Testing, Statistical Modeling, Logistic Regression, Time Series Forecasting, Spectral Analysis, State-Space Modeling, Clustering
- **Technical:** *R*, *Python*, *SQL*, Power BI, Tableau, Excel, Git/GitHub, Snowflake, AWS, ETL/ELT
- **Libraries:** pandas, numpy, statsmodels, sklearn, DoWhy, causalML, tensorflow, pytorch, tidyverse, plotly, matplotlib, Shiny

EXPERIENCE

Iridium Satellite Communications

Data Science Intern

May 2025 - May 2026

- Saved ~\$25k annually by reducing flight data pulls by 75%; engineered an ML model predicting global flight routes using HDBSCAN clustering and intent-based learning (Python)
- Discovered 200+ day subscriber provisioning lags across product lines through deep-dive survival analysis research (R), translating timing delays into activation probability predictions that shaped business and supply chain planning
- Consolidated message performance data across satellite, ground, aircraft, and partner systems into centralized measurement dashboards, automating reporting to demonstrate 99%+ delivery timing compliance for FAA flight trials

Aviation Business Development Intern

May 2024 - August 2024

- Built ETL data pipelines using APIs and CTE-based, long-form SQL with geospatial time/distance matching to extract and unify subscriber and flight data from 76k aircraft and 6k devices across 17M flights into unified analytical schema
- Built company's first interactive, online route map tracking key aviation service metrics across global markets
- Delivered subscriber acquisition and pricing insights to internal and partner stakeholders through Power BI & Microstrategy

Biokind Analytics

Lead Data Analyst

January 2025 - December 2025

- Managed project delivery and guided team on translating complex statistical findings for non-technical stakeholders, led a project involving time series analysis of 10-year donor history to inform program decisions
- Provided statistical consulting to healthcare humanitarian nonprofits, including survival analysis, nonparametric correlation, logistic regression for risk factors, associations for relapse of acute malnutrition in children

University of Virginia

Statistics Teaching Assistant

August 2025 - May 2026

- Explained statistical simulation techniques to over 80 students across statistics and non-statistics majors in the Data to Knowledge course; topics included the bootstrap and Monte Carlo, sampling and estimation, and coding in R

UVA Sports Analytics and Statistics Laboratory (SASL)

Lab Intern

September 2023 - December 2025

- Team Lead UVA's [NFL Big Data Bowl](#) team; led graduate researchers, metrics adopted within lab's research portfolio

Last Call

Band Leader/Percussionist

August 2022 - May 2026

- Performed for a packed audience of 7k+ at John Paul Jones Arena, opening for multi-platinum band Neon Trees; leading Last Call, a group featuring a *Voice* contestant and nationally recognized touring/jazz musician

PROJECTS

Phase II Clinical Trial Protocol

February 2026 - May 2026

- Designed a Phase II obesity-drug RCT protocol (n=68, 4-arm 1:1:1:1 randomization) grounded in recent Phase II/III obesity-trial literature, with ANCOVA-based primary analysis and Holm-adjusted pairwise comparisons

Drumming Styles: A Distributional Feature Analysis

August 2025 - December 2025

- Engineered 32-D feature space from 446,312 MIDI drum hits across 10 drummers; applied Gaussian KDE, PCA (63.8% variance), and 3-state Hidden Markov Models revealing drummer-specific state persistence patterns (range: 0.25-0.70)

Album Covers and Billboard Success

August 2025 - December 2025

- Engineered 70-D visual feature space from 2,084 album covers using scikit-image and YOLO for face detection; achieved 0.87 ROC-AUC with L1-regularized logistic regression predicting Billboard 200 success
- Identified LBP texture features, edge density, and human presence as strongest predictors of chart performance

Impact of Recession on Housing Starts Forecasting

August 2025 - December 2025

- Demonstrated regime-dependent forecast failure across the 2008 Financial Crisis using SARIMA(1,0,1)(1,0,1)[12]; pre-crisis models systematically overpredicted post-crisis recovery; spectral analysis revealed 8 to 10.4-year cycle shift